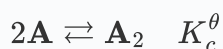


Question

Compound **A** can undergo dimerization to form A_2 , and this dimerization reaction constitutes the following equilibrium:



The equilibrium constant of the reaction can be measured by polarimetry, which involves determining the specific optical rotation $[\alpha]_D$ of the solution when the reaction reaches equilibrium and then calculating the equilibrium constant K_c^θ . It is known that when the reaction reaches equilibrium, $[\alpha]_D$ satisfies the following relationship:

$$[\alpha]_D = x_1^2[\alpha]_1 + x_2^2[\alpha]_2$$

where x_1 and x_2 are the mole fractions of the corresponding species, and $[\alpha]_1$ and $[\alpha]_2$ are the specific optical rotations of the corresponding species when they exist independently.

1. The structure of compound **A** is shown in Figure 1. The task here is to deduce the structure of its dimer A_2 .

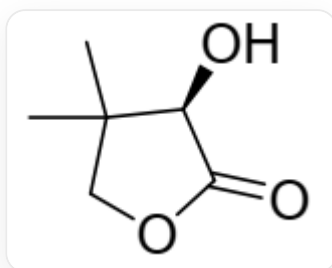


Fig. 1. The molecule is described by SMILES as: CC1(C)COC(=O)[C@@H]1O

2. At room temperature, a solution of compound **A** with a total concentration of 8.0mmol/L in CCl_4 solvent reaches equilibrium, and the specific optical rotation of the system is measured as

$$[\alpha]_{\text{D}} = -6 \text{ deg dm}^{-1} \text{ g}^{-1} \text{ mL}.$$

The specific optical rotations of compound **A** and dimer **A**₂ when they exist independently are

$$[\alpha]_1 = -1 \text{ deg dm}^{-1} \text{ g}^{-1} \text{ mL}$$

$$[\alpha]_2 = -203$$

$$\text{deg dm}^{-1} \text{ g}^{-1} \text{ mL}$$

Ignoring other possible species in the system, calculate the equilibrium constant K_c° for this dimerization reaction.

The following statements are given:

1. A covalent bond is formed during the dimerization of **A**.
2. The dimer **A**₂ does not contain a C_2 symmetry axis.
3. The equilibrium constant for the dimerization reaction is $K_c^\circ = 33.5$.
4. The equilibrium constant for the dimerization reaction is $K_c^\circ = 3.33$.
5. The equilibrium constant for the dimerization reaction is $K_c^\circ = 28.7$.

Which of the statements above are correct?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 1,3

G. 1,4

H. 1,5

I. 2,3

J. 2,4

K. 2,5

L. 1,2

M. 1,2,3

N. 1,2,4

O. 1,2,5

P. 2,3,4

Q. 2, 3, 5

R. 3,4,5

S. 1, 2, 3, 4

T. 1, 2, 4, 5

U. 2, 3, 4, 5

V. 1,3,4,5

W. 1, 2, 3, 4, 5

X. None of the above options is correct

Answer

Correct Answer: **C**

Detailed Explanation

Observing the structure of compound **A**, it contains:

Hydroxyl group ($-\text{OH}$)

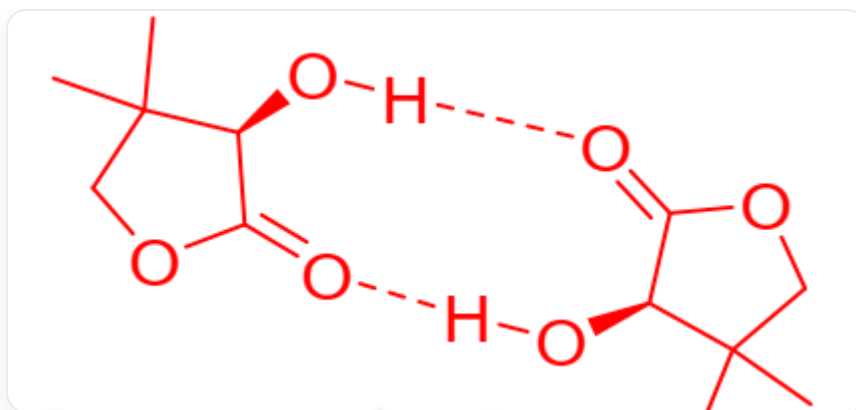
Carbonyl group ($\text{C}=\text{O}$)

These two functional groups can interact through hydrogen bonding. The dimer A_2 is most likely formed via intermolecular hydrogen bonds, where the ($-\text{OH}$) of one molecule forms a hydrogen bond with the ($\text{C}=\text{O}$) of another molecule, resulting in a cyclic dimer structure as shown below. The dimerization process does not involve covalent bond formation, so statement 1 is incorrect. The two monomers are connected by hydrogen bonds, forming two $\text{C}=\text{O} \cdots \text{H}-\text{O}$ hydrogen bonds in total. This structure exhibits C_2 symmetry, so statement 2 is incorrect. (An alternative form involves intermolecular transesterification to form a six-membered ring, with the product structure represented in SMILES as: CC(CO)([C@@H](OC([C@H](O1)C(CO)(C)C)=O)C1=O)C. However, this process is entropically unfavorable compared to intramolecular formation of a five-membered ring ester bond.)

CHECKPOINT

1 PTS

Intermolecular transesterification to form a six-membered ring dimer is entropically unfavorable



The image shows the structure of the dimer $\mathbf{A}_2$, where the monomer $\mathbf{A}$ remains unchanged. $\mathbf{A}$ is described in SMILES as: CC1(C)COC(=O)[C@@H]1O. The two monomers are connected by hydrogen bonds between the hydroxyl (--OH) and carbonyl ($\text{C}=\text{O}$) groups, forming two $\text{C}=\text{O} \cdots \text{H}-\text{O}$ hydrogen bonds. The image is a schematic diagram of the chemical structure, featuring two mirror-image molecular structures interacting via two dashed lines. Each molecular structure centers on a five-membered ring containing two oxygen atoms, denoted by the letter "O." One of the ring oxygen atoms is adjacent to a carbonyl group ($\text{C}=\text{O}$), forming a lactone structure, with the carbonyl oxygen also labeled "O." On each five-membered ring, a carbon atom is connected to a hydroxyl group via a solid wedge bond, where the hydroxyl consists of an oxygen atom "O" and a hydrogen atom "H." Another carbon atom on the ring is bonded to two methyl groups, represented by short lines. The interaction at the center of the image is depicted by two dashed lines: one between the hydroxyl hydrogen "H" of the left molecule and the carbonyl oxygen "O" of the right molecule, and another between the hydroxyl hydrogen "H" of the right molecule and the carbonyl oxygen "O" of the left molecule. All chemical bonds in the diagram are represented by solid lines, except for the dashed lines indicating intermolecular interactions, wedge bonds for stereochemistry, and double lines for $\text{C}=\text{O}$ bonds.

CHECKPOINT

1 PTS

$\mathbf{A}$ dimerizes via hydrogen bonding

CHECKPOINT

1 PTS

The dimer exhibits C_2 symmetry.

Let the mole fractions of **A** and **A**₂ be x_1 and x_2 , respectively, where $x_1 + x_2 = 1$.

Substituting the known specific rotation data into $[\alpha]_D = x_1^2[\alpha]_1 + x_2^2[\alpha]_2$ yields: $-x_1^2 - 203x_2^2 = -6$.

Solving these two equations simultaneously gives $x_1 = 0.838$ and $x_2 = 0.161$.

CHECKPOINT

1 PTS

Mole fraction of **A**=0.838, Mole fraction of **A**₂=0.161

Given $c(\mathbf{A}) + 2c(\mathbf{A}_2) = 0.0080 \text{ mol/L}$, solving simultaneously yields $c(\mathbf{A}) = 5.78 \times 10^{-3} \text{ mol/L}$.

Thus, $K_c^\theta = c(\mathbf{A}_2) \times c^\theta / [c(\mathbf{A})]^2 = (1.12 \times 10^{-3}) / (5.78 \times 10^{-3})^2 = 33.5$.

Statement 3 is correct, while statements 4 and 5 are incorrect, making option C the correct choice.

CHECKPOINT

1 PTS

Equilibrium constant $K_c^\circ = 33.5$